

# XGeom

## Parametric geometry for aero and marine shape optimization

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### ABOUT

Working document for XGeom. Covers scope, approach, status, plan and open questions, and is updated as the work progresses. Changes are recorded in the decision log and the change log at the end.

## 1 Goal

XGeom is a parametric geometry tool for shape optimization of lifting surfaces (wings, rudders, propeller and turbine blades) and, later, hulls. Required functions:

### Build

Generate geometry from a small set of parameters: a section shape plus spanwise laws for chord, twist and offsets. Output surface points and a B-rep solid.

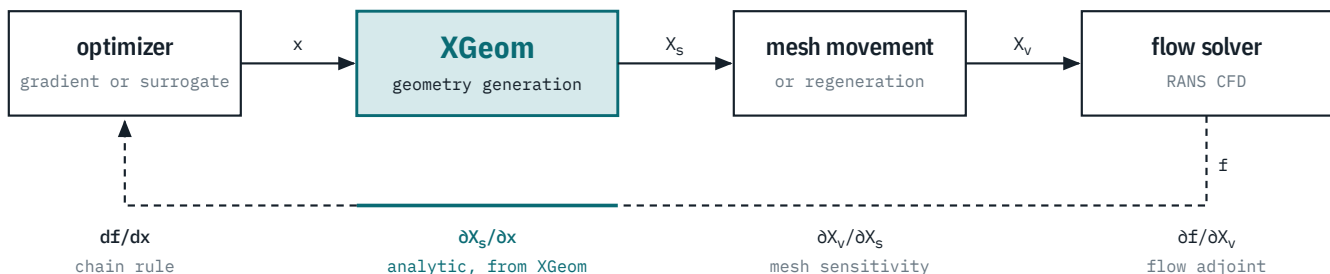
### Fit

Recover the parameters from an existing STEP or IGES model by cutting it into sections.

### Differentiate

Return the analytic Jacobian  $\partial X_s / \partial x$  of the surface points with respect to every free parameter, for adjoint or surrogate-based optimization.

The analytic Jacobian is the main requirement. In a gradient-based loop,  $df/dx = \partial f / \partial X_v \cdot \partial X_v / \partial X_s \cdot \partial X_s / \partial x$  (Samareh 2001). The flow adjoint and the mesh movement provide the first two factors; the geometry tool must provide the last one (Fig. 1).



**Fig. 1.** XGeom in the optimization loop. Solid arrows: forward evaluation. Dashed arrows: gradient.

## Existing tools

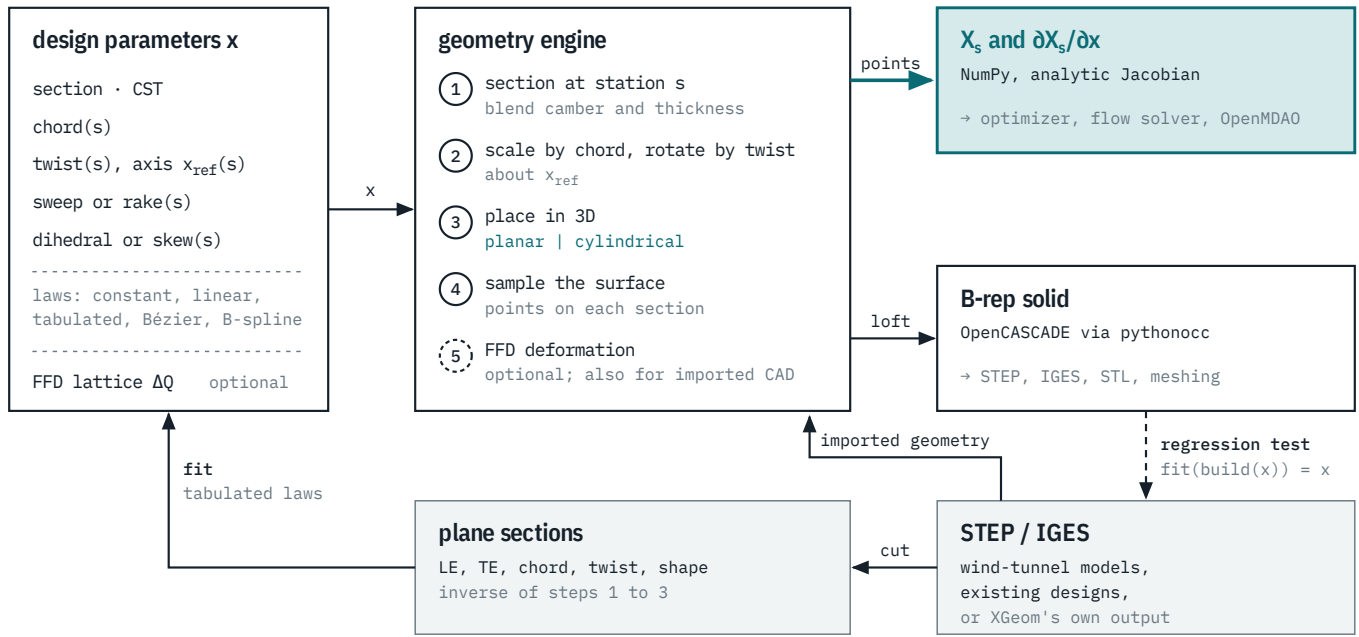
pyGeo (Hajdik et al. 2023) is the closest open-source option. Its FFD is analytic but requires an existing geometry, and its constructive options (OpenVSP, ESP) reach the optimizer through finite differences (Table 1). ESP computes analytic sensitivities for many of its own CAD operations; review it before the design is fixed.

**Table 1.** Options in pyGeo, as reported by Hajdik et al. (2023), compared with the XGeom target.

| APPROACH                     | TYPE                       | 3D      | DERIVATIVES                 | STARTS FROM                                      |
|------------------------------|----------------------------|---------|-----------------------------|--------------------------------------------------|
| FFD in pyGeo                 | deformative                | yes     | analytic, complex step      | existing geometry                                |
| CST in pyGeo                 | constructive               | 2D only | analytic                    | parameters                                       |
| OpenVSP or ESP through pyGeo | constructive               | yes     | parallel finite differences | parameters                                       |
| XGeom (target)               | constructive, optional FFD | yes     | analytic                    | parameters, a fit to CAD, or any CAD through FFD |

## 2 Approach

One engine builds all lifting surfaces in five steps (Fig. 2). Outputs: surface points with their analytic Jacobian for the optimizer, and a B-rep solid for CAD export and meshing. Only the points are differentiated, so OpenCASCADE stays out of the gradient and is confined to one module. Fitting runs the same steps in reverse;  $\text{fit}(\text{build}(x)) = x$  is the main regression test.



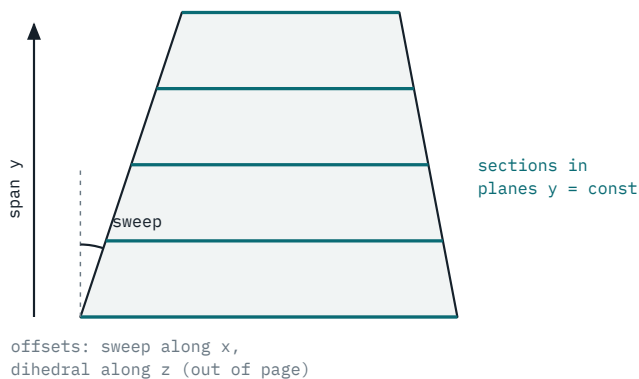
**Fig. 2.** Planned architecture. Build along the top, fit along the bottom; the highlighted path is differentiated. Step 5 (FFD) is optional and also applies to imported CAD.

## Section placement

Wings and blades differ only in step 3 (Fig. 3). Wing-type surfaces place sections in parallel planes; propellers and axial turbomachinery wrap them onto cylinders. Everything else is which spanwise laws are active. Presets (wing, rudder, fin, propeller, turbine blade) set defaults only, since the wind-tunnel rudder already carries 4.0° of leading-edge sweep aft and 8.1° of trailing-edge sweep forward.

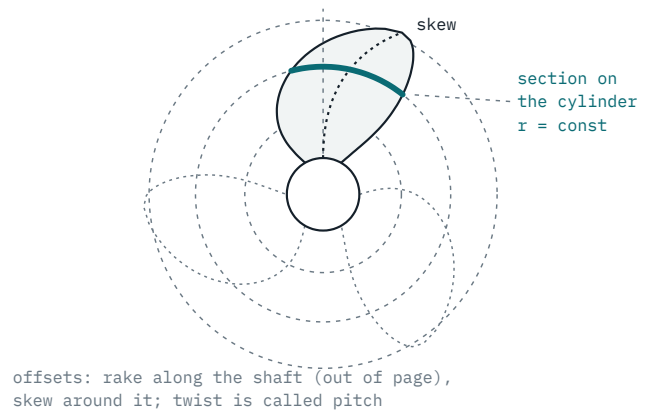
### planar

wings, rudders, fins, keels, wind-turbine blades



### cylindrical

marine propellers, axial turbomachinery



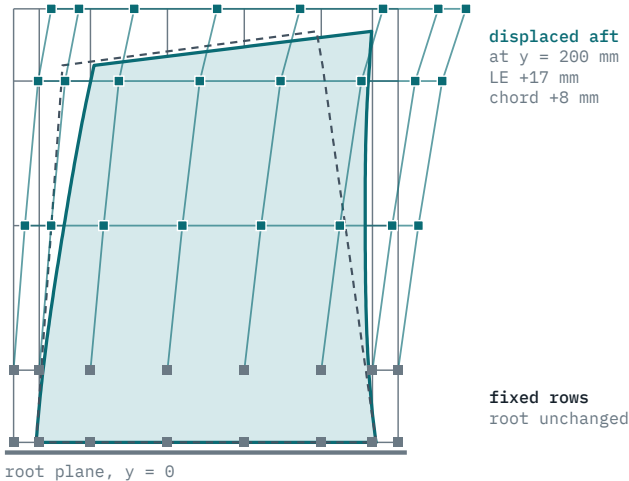
**Fig. 3.** Planar and cylindrical placement (schematic). Highlighted lines are sections.

## FFD (optional step 5)

Free-form deformation (Sederberg and Parry 1986): moving the control points of a lattice deforms everything inside it. Applies to hulls and to imported CAD with no parametric model, including the rudder tip not covered by the section cuts. The deformation is linear in the control-point moves, so its Jacobian is a fixed sparse matrix computed once. Fig. 4 shows a first test on the rudder.

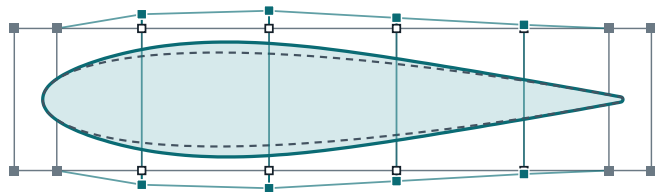
(a) Planform

lattice  $8 \times 5$ , cubic in  $x$  and  $y$



(b) Section B-B,  $y = 80$  mm

lattice  $8 \times 2$ , cubic in  $x$ , linear in  $z$



linked rows:  $\Delta z(\text{upper}) = -\Delta z(\text{lower})$   
LE and TE columns fixed; section remains symmetric  
t/c 0.162  $\rightarrow$  0.198

■ fixed control point      ■ displaced control point  
-- original geometry      □ deformed geometry  
— original lattice      — deformed lattice

**Fig. 4.** FFD computed on the extracted rudder with B-spline lattices. Control points sit at the Greville abscissae, so the undeformed lattice reproduces the shape exactly. (a) Root rows fixed and upper rows moved aft: more sweep and a longer tip chord, root unchanged; tip outline schematic. (b) Upper and lower rows moved by equal and opposite amounts: the section at  $y = 80$  mm thickens and stays symmetric.

## Target interface

Not implemented yet.

```
rudder = Rudder.fit("Wind_Tunnel_Rudder.stp") # fit parameters to the CAD model
rudder.twist = Bezier(4, axis=0.5) # free twist about mid-chord
x = rudder.design_vector() # free parameters and bounds
Xs = rudder.surface(x) # surface points
J = rudder.jacobian(x) # analytic Jacobian dXs/dx
rudder.to_step(x, "rudder_opt.stp") # B-rep export
```

OpenMDAO wrapper: compute calls surface, compute\_partials calls jacobian.

### 3 Status

in progress

working

done (MSc)

not started

- Literature review.

Rudder extraction.

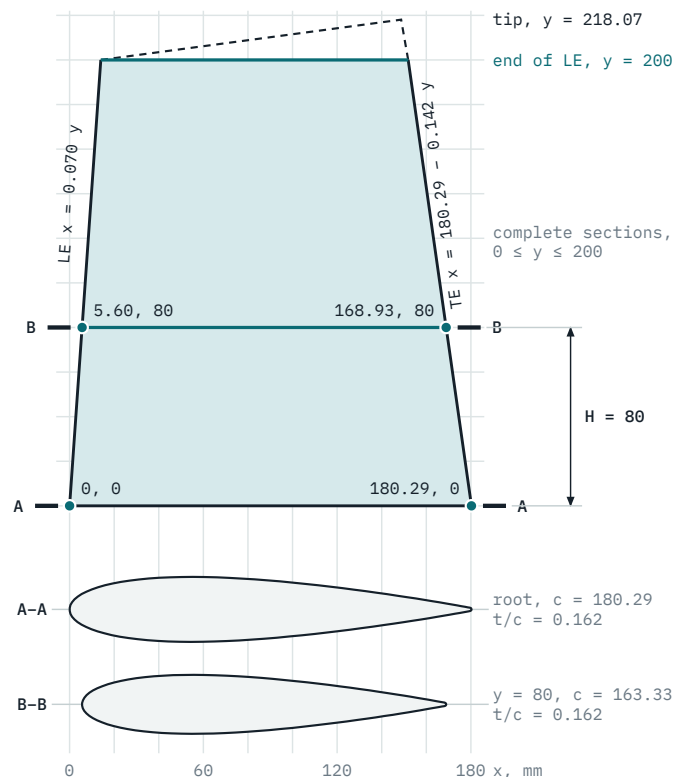
Propeller.

Code base.
- 7 of 27 papers read, plus the OpenMDAO paper (section 7).

Fit side of Phase 0 working on the wind-tunnel rudder (Fig. 5).

Parametric propeller generation exists from the MSc work. Port to the cylindrical placement in P3.

Waits on the section-basis decision (section 5).



**Fig. 5.** Extracted wind-tunnel rudder at true scale, with sections A–A (root) and B–B ( $y = 80$  mm). Grid 20 mm. Shaded: where horizontal cuts give complete sections. The tip above  $y = 200$  mm (dashed) is schematic and not covered yet.

#### FRAME

Origin at the root leading edge,  $x$  chordwise,  $y$  spanwise,  $z$  thickness, in mm. Reframed by a  $180^\circ$  rotation about the span axis (det +1); a mirror would flip the normals.

#### EDGES

LE  $x = 0.070 y$ , TE  $x = 180.29 - 0.142 y$ , both straight up to  $y = 200$  mm, where the tip blend starts.

#### ACCURACY

Sections lie on the B-rep to  $5 \times 10^{-7}$  mm; chords match an independent STL slice to  $1 \times 10^{-4}$  mm.

#### SECTION

Symmetric, 16.2 % thick at 30 % chord (NACA 0016 family), small rounded trailing edge.

## OUTPUT

Any number of sections up to  $y = 200$  mm as .dat files in Selig order; 200 sections take about 30 s.

## 4 Plan

Phases in planned order. Each phase closes when its test passes. The order may change as the open questions are resolved.



P5

## CFD integration planned

Add watertight export, mesh-ready cases, and adapters for OpenMDAO and flow solvers.

**DONE** WHEN a design vector goes in and a mesh-ready case comes out with its Jacobian.

### Next steps

1. Fit the rudder root section with CST at increasing Bernstein order, down to the one-count tolerance of Masters et al. (about 9  $\mu\text{m}$  on the 180 mm chord, 4.5  $\mu\text{m}$  near the leading edge).
2. Read Hicken and Zingg (2010) and Lee, Koo and Zingg (2017) for the solver hand-over question.
3. Set up the repository and move the extraction code into the sectioning module.

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## 5 Open questions

Undecided items, with the current direction and what resolves each.

### Q1 Which section basis first?

**Current direction:** CST with the leading-edge term (Masters et al. 2017). **Resolve by:** the CST fits of the rudder root section.

### Q2 How does XGeom hand geometry to the flow solver?

Many optimization codes carry their own geometry control tied to mesh movement, such as B-spline volumes (Hicken and Zingg 2010) or FFD in pyGeo. Options: generate the baseline once, or stay in the loop and supply  $\partial X_s / \partial x$  at every iteration. **Current direction:** undecided. **Resolve by:** the two readings in the next steps.

### Q3 Where does dimensionality reduction happen?

At 20 to 25 variables per section, a 3D surface quickly outgrows a surrogate. **Current direction:** inside XGeom for the parts linear in their variables (CST coefficients, FFD), where the Karhunen–Loève modes (Diez et al. 2015) come from one SVD of the Jacobian. Undecided for the planform variables.

### Q4 Is a marine section basis worth building?

SVD modes were the most efficient basis in Masters et al. (2017), but only on the library they were trained on. No rudder or hydrofoil library was tested, so modes trained on marine sections could be a contribution. **Current direction:** revisit after the CST fits.

### Q5 How close is ESP to the XGeom target?

ESP has analytic sensitivities for many of its CAD operations. **Resolve by:** an ESP sensitivity paper, read before the architecture is fixed.

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## 6 Decisions

| DATE        | DECISION                                                                                                                                 |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 14 Sep 2026 | Use one engine with thin presets (wing, rudder, blade, and so on), not a separate class per geometry type.                               |
| 14 Sep 2026 | Implement both build and fit in the first version; $\text{fit}(\text{build}(x)) = x$ is the main test.                                   |
| 14 Sep 2026 | Make the design vector and its Jacobian part of the core from Phase 1.                                                                   |
| 14 Sep 2026 | Lifting-surface frame: origin at the root leading edge, x chordwise, y spanwise, z thickness. Rotate incoming CAD into it; never mirror. |
| 14 Sep 2026 | Complete the key literature before writing any code structure (Phase -1).                                                                |
| 24 Sep 2026 | Add FFD as an optional deformation step (P1b).                                                                                           |
| 24 Sep 2026 | Keep the flow-solver interface general for now.                                                                                          |
| 24 Sep 2026 | Port the propeller from the MSc code; do not rewrite it.                                                                                 |

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## 7 Reading log

| PAPER               | TAKEAWAY                                                                                                                                                                                                                                                                                |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Samareh 2001        | Survey of parametrization methods. The geometry tool is responsible for $\partial X_s / \partial x$ in the gradient chain.                                                                                                                                                              |
| Masters et al. 2017 | CST with the leading-edge term is the best first section basis. 20 to 25 variables per section cover all 2,174 test airfoils to Kulfan's tolerance; the leading-edge term gives 80 % coverage with 16 variables instead of 22. One-count accuracy needs 38 to 66 variables per section. |
| Hajdik et al. 2023  | pyGeo. FFD is analytic; OpenVSP and ESP go through finite differences. The common interface and constraint handling are worth copying.                                                                                                                                                  |
| Kulfan 2008         | CST. The same functions describe a 3D wing in 19 variables and a nacelle in 15, so lifting surfaces and bodies can share one section representation.                                                                                                                                    |
| Gray et al. 2019    | OpenMDAO. Each component supplies its own partial derivatives, so an XGeom component is a thin adapter.                                                                                                                                                                                 |
| Park et al. 2021    | Twisted rudder. Twist is applied about the mid-chord and follows the propeller-induced inflow, so the twist axis must be a parameter.                                                                                                                                                   |
| Cheng et al. 2024   | Hull, propeller and rudder, all fully parametric. 65 hull parameters defined, 8 used as design variables: keep the model separate from the design space.                                                                                                                                |

| PAPER           | TAKEAWAY                                                                                         |
|-----------------|--------------------------------------------------------------------------------------------------|
| Han et al. 2012 | Review of hull form optimization. Hull curves are fitted with fairness as an explicit objective. |

**Next to read:** Hicken and Zingg (2010), Lee, Koo and Zingg (2017), Sederberg and Parry (1986), Diez et al. (2015), and an ESP sensitivity paper.

## References

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Diez, M., Campana, E. F., Stern, F. (2015). Design-space dimensionality reduction in shape optimization by Karhunen–Loève expansion. *Comput. Methods Appl. Mech. Eng.* 283, 1525–1544.

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Hicken, J. E., Zingg, D. W. (2010). Aerodynamic optimization algorithm with integrated geometry parameterization and mesh movement. *AIAA J.* 48(2), 400–413.

Kulfan, B. M. (2008). Universal parametric geometry representation method. *J. Aircraft* 45(1), 142–158.

Lee, C., Koo, D., Zingg, D. W. (2017). Comparison of B-spline surface and free-form deformation geometry control for aerodynamic optimization. *AIAA J.* 55(1), 228–240.

Masters, D. A., Taylor, N. J., Rendall, T. C. S., Allen, C. B., Poole, D. J. (2017). Geometric comparison of aerofoil shape parameterization methods. *AIAA J.* 55(5), 1575–1589.

Park, I., Paik, B., Ahn, J., Kim, J. (2021). The prediction of the performance of a twisted rudder. *Appl. Sci.* 11(15), 7098.

Samareh, J. A. (2001). Survey of shape parameterization techniques for high-fidelity multidisciplinary shape optimization. *AIAA J.* 39(5), 877–884.

Sederberg, T. W., Parry, S. R. (1986). Free-form deformation of solid geometric models. *Proc. SIGGRAPH '86*, 151–160.

## Change log

- 25 Sep 2026 Restructured as a working document: status, open questions, decision and reading logs. FFD figure added. Propeller marked as done (MSc).
- 24 Sep 2026 First version, used to present the concept and initial architecture.